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Silk Road glass in Xinjiang, China: chemical compositional analysis and interpretation using a high-resolution portable XRF spectrometer

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ABSTRACT

In this paper we explore the major, minor and some trace elements of sixty-five glass beads from eighteen oasis sites both north and south of the Taklamakan Desert, the ancient centre of the Silk Routes linking East and West; the samples date from the Warring States period (475–2211 BCE) to the Tang Dynasty (618–907 CE). Using a high-resolution portable XRF spectrometer, with special attention to issues of surface weathering and sample preparation, we obtained quantitative chemical compositional data that, when combined with archaeological context and the technology of bead manufacture, provide the basis for an understanding of both changes in glass type over time, and possible source areas for the Xinjiang glasses. While glass beads at the early sites (1st c. BCE to 4th c. CE) were coming primarily from Kushan-controlled territories in northern India, Pakistan and Afghanistan, with a small contribution from central China, most of the later glass (4th–10th c. CE) is similar to that produced in the Sasanian Empire at such sites as Veh Ardashir in central Iraq. These changes most likely reflect changing cultural and economic conditions in the Silk Road towns.

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1. Introduction

The Xinjiang Uygur Autonomous Region (Xinjiang) in northwest China has been a crossroads of cultural diversity and complexity from the Chinese Bronze Age (ca 2000–1000 BCE) to the present day (Chen, 1990). Even before the Western Han Dynasty (202 BCE–25 CE), people in Xinjiang participated in mutual exchanges of culture, technology and trade goods with the West and with central China. While Chinese silk is perhaps the most familiar of the traded commodities, glass, in the form of ornaments and vessels, was very important in this exchange, and glass objects from the Roman and Sasanian Empires and Central Asia became treasures in China, Korea and Japan (An, 1984; Gan, 2009; Lee, 2010; Taniichi, 2010). In recent years, Chinese archaeologists have found many glass artifacts of different historical periods in Xinjiang. These glass artifacts, including beads, tubes, vessels, pendants and other decorative goods, have been found at sites dated from the late Western Zhou Dynasty (*c.* 1046–771 BCE) to the Song Dynasty (960–1276 CE) (Zhang, 2007; Yu, 2010). While the earliest glass vessels appear during the Han Dynasty (202 BCE–220 CE), glass and faience beads dated from the Western Zhou Dynasty to the Warring States Period (475–221 BCE) have been excavated in Xinjiang at such sites as the Qun-ba-ke cemetery (*c.* 955–680 BCE) in Luntai county, and the Xia-ban-di cemetery (*c.* 1000–500 BCE) in Taxkorgan county.

As early as the late Spring and Autumn Period (770–476 BCE), glassmaking technology passed eastwards along the Silk Road, stimulating indigenous Chinese glass production (Gan et al., 2006, 2008). From the Warring States Period (475-221 BCE) to the Western Han Dynasty (202 BCE–25 CE), Chinese craftsmen working in central and southern China had developed new glass recipes incorporating high levels of lead (Pb) and barium (Ba), perhaps in an effort to reproduce the visual qualities of jade (Li et al., 2003, 2005; Zhang et al., 2004; Gan et al., 2010). Glass beads made from lead-barium glass, either monochrome or decorated with compound eyes, have been found not only in central China, but also in late Warring States to Han Dynasty contexts in Hami and Hetian, Xinjiang (Gan et al., 2006).

Early western glass bead imports in Xinjiang include those excavated from both Baicheng county and Tacheng county that were dated to 1000–500 BCE. In addition to their soda lime glass compositions, some of these beads with yellow colour contained lead and antimony (Gan et al., 2005), a possible further indication of a Western origin. In central China, the earliest soda lime glass eye

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beads date from the late Spring and Autumn period (770–476 BCE) to the early Warring States period (475–221 BCE), with examples found at such places as Xichuan in Henan province and Sui county in Hubei province (Li et al., 2005; Gan et al., 2008).

Of the sites so far excavated in Xinjiang, only the Wa-shi-xia site in Ruoqiang county, Xinjiang, dated from the Song to early Yuan Dynasty (ca 10th–14th c. CE), revealed evidence for glassmaking (Zhang, 1986). Archaeologists excavated more than ten fragments of finished glass bottles and jars, and many unfinished glass chunks with different colours, indicating at least secondary glassmaking-the production of finished objects. In addition, large amounts of charcoal used as fuel were found. No clear archaeological evidence shows that craftsmen in Xinjiang could make glass locally before the Song Dynasty, suggesting that most of the glass beads analyzed in the present paper were imported from places outside of Xinjiang.

The number of previous reports on chemical compositions of early glass in Xinjiang is quite limited, with recent studies including those by Robert Brill (2009) of glass samples collected by early European explorers and archaeologists, and by Yi-xian Lin (2010). From this exploratory work the most useful result may be Brill's observation that plant ash glass with potash levels greater than 4 wt% suggests a Central Asian provenance. In addition, Brill identified glass groups found in Central Asia with high potash and low, medium or high alumina among the samples he studied. Lin expanded on Brill's work in her study of 36 glass beads from the Niya site (2nd c. BCE–4th c. CE), finding, among other compositional types, three groups of plant ash soda lime glass separated primarily on the basis of alumina (Al_2O_3) content (Lin, 2010). Most of Lin's samples fell in the moderate (2–4 wt%) alumina region, and these had potash values (not shown) greater than 4.0 wt%. In addition to these samples, classified as Central Asian, there were six beads, including a mosaic eye bead very similar to our XJ-40-2 (see below), with higher alumina, that she links to glass from the Bara site in Pakistan (Dussubieux and Gratuze, 2003). While these recently published studies are extremely interesting and suggestive, still lacking is a broad survey of glass finds from Xinjiang that might help us to understand the complex and changing nature of both Silk Road glass and Silk Road trade in general.

Our goals for the study reported here are two-fold. The first is to provide evidence for the morphological and chemical compositional types of glass beads recovered from Xinjiang sites representing both geographical and chronological divisions. The second goal is to evaluate the usefulness of a high-resolution portable x-ray fluorescence (*p*XRF) spectrometer to provide glass compositional data of a quality that permits relatively fine-grained interpretation. Although the analyses reported here were performed in our laboratory in Shanghai, we may now, based on the success of this pilot study, continue our *p*XRF work at several archaeological sites. Presented here are the first published analyses from the use of such equipment in China, and, in addition, the first use of this high-resolution *p*XRF system by a group other than that of the developers at the Tokyo University of Science (Nakai et al., 2005; Tantrakarn et al., 2009; Kato et al., 2010).

The immediate archaeological questions are what compositional types of glass were important at these Xinjiang sites, and what were the possible source areas for the samples to be analyzed? We hope to be able to interpret these results in terms of political, economic and cultural patterns in the lives of people in the oasis settlements of Xinjiang both north and south of the Taklamakan Desert. In addition, we are interested in how best to combine the goals of chemical analysis of a broad range of glasses, on site if possible, with the use of a minimally destructive technique. For this, we have explored both the instrumental and the sample-related issues that have limited the usefulness of *p*XRF for glass study.

2. Materials

Sixty-five glass beads and two glass bracelet fragments either excavated or gathered from eighteen Xinjiang archaeological sites of various dates were analyzed by *p*XRF. Fig. 1 is a map of Xinjiang showing the locations of the oasis settlements around the Taklamakan Desert where the glass samples were found, and Table 1 lists all of the samples, along with the archaeological sites where the beads were found and their approximate dates. The archaeological contexts of the excavation sites have been published elsewhere (Qi and Wang, 2008) and, for ease of reference, a summary of this information is available as an online supplement to this article. In addition, both the general morphology and colour of the samples are indicated. Photographs of many of the beads are included in

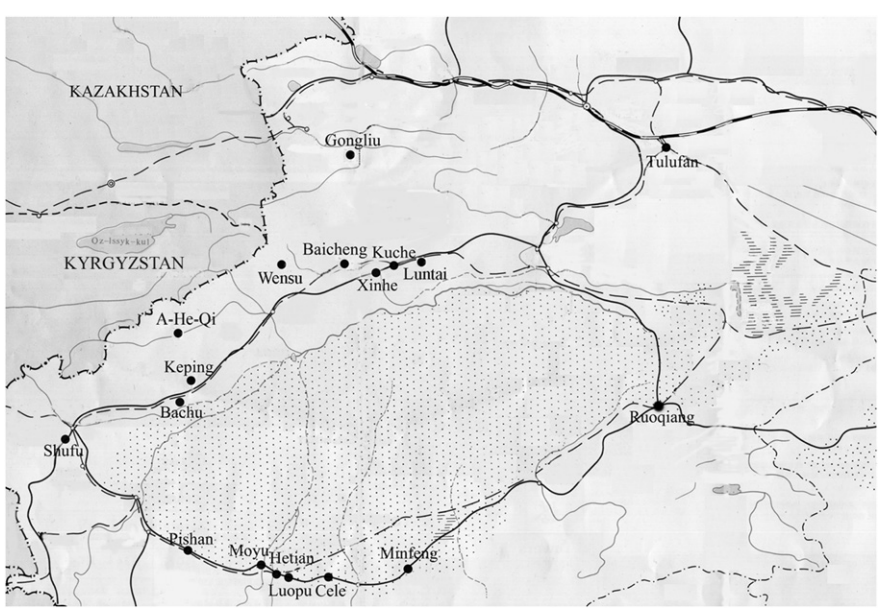

Fig. 1. Map showing the locations of the historical sites mentioned in the text.

Table 1
Glass samples by site, date, morphology and colour. All samples are beads unless otherwise noted. Colour abbreviations: B (blue), Bn (brown), G (green), Bk (black), Cls (colourless), R (red), Y (yellow), W (white), V (violet); l and d indicate light and dark; prefix b, g and p indicate bluish, greenish and purplish.

Sample no.	Site	Date	Morphology	Colour
XJ-9-1	Da-Ma Gorge site, Cele county	4–8c CE	Monochrome drawn	gB
XJ-9-2	Da-Ma Gorge site, Cele county	4–8c CE	Monochrome wound (?)	Bn
XJ-9-3	Da-Ma Gorge site, Cele county	4–8c CE	Bracelet fragment	gB
XJ-9-4	Da-Ma Gorge site, Cele county	4–8c CE	Bracelet fragment	Amber
XJ-10-1	Da-Ma Gorge site, Cele county	4–8c CE	Monochrome wound (?)	gB
XJ-10-2	Da-Ma Gorge site, Cele county	4–8c CE	Monochrome wound (?)	G
XJ-10-3	Da-Ma Gorge site, Cele county	4–8c CE	Monochrome drawn (?)	bG
XJ-10-4	Da-Ma Gorge site, Cele county	4–8c CE	Monochrome	gB
XJ-10-5	Da-Ma Gorge site, Cele county	4–8c CE	Monochrome drawn	pB
XJ-10-6	Da-Ma Gorge site, Cele county	4–8c CE	Monochrome wound (?)	G
XJ-10-7	Da-Ma Gorge site, Cele county	4–8c CE	Monochrome drawn (?)	Bn
XJ-10-8	Da-Ma Gorge site, Cele county	4–8c CE	Faceted, possibly lapidary worked	bG
XJ05-3-1	Niya site, Minfeng county	2c BCE–4c CE	Trailed decoration wound	Bk
XJ05-3-2	Niya site, Minfeng county	2c BCE–4c CE	Segmented gold-glass drawn	Cls
XJ05-3-3	Niya site, Minfeng county	2c BCE–4c CE	Trailed decoration wound	Bk
XJ05-3-4	Niya site, Minfeng county	2c BCE–4c CE	Trailed *vs* folded decoration	Bk
XJ05-3-5	Niya site, Minfeng county	2c BCE–4c CE	Folded with circumferential trail	Bk, W, R
XJ05-3-6	Niya site, Minfeng county	2c BCE–4c CE	Folded with circumferential trail	Bk, W
XJ05-3-7	Niya site, Minfeng county	2c BCE–4c CE	Stratified eye	gB, Bk, W
XJ05-3-8	Niya site, Minfeng county	2c BCE–4c CE	Trailed *vs* folded decoration	Bk, W
XJ-8-1	Niya site, Minfeng county	2c BCE–4c CE	Faceted, drawn(?) and paddled	gB
XJ-8-2	Niya site, Minfeng county	2c BCE–4c CE	Segmented drawn	gB
XJ-16-1	Qiong-Ti-Mu site, Keping county	6–8c CE	Monochrome drawn	G opaqe
XJ-16-2	Qiong-Ti-Mu site, Keping county	6–8c CE	Monochrome wound (?)	B
XJ-16-3	Qiong-Ti-Mu site, Keping county	6–8c CE	Monochrome wound	lG
XJ-16-5	Qiong-Ti-Mu site, Keping county	6–8c CE	Monochrome drawn (?)	lB
XJ-16-6	Qiong-Ti-Mu site, Keping county	6–8c CE	Zone	gB, Bn, W
XJ-16-7	Qiong-Ti-Mu site, Keping county	6–8c CE	Monochrome wound (?)	B
XJ-16-8	Qiong-Ti-Mu site, Keping county	6–8c CE	Monochrome	gB
XJ-16-9	Qiong-Ti-Mu site, Keping county	6–8c CE	Monochrome wound (?)	G
XJ-16-10	Qiong-Ti-Mu site, Keping county	6–8c CE	Trailed to imitate mosaic	Bn, Y
XJ-16-11	Qiong-Ti-Mu site, Keping county	6–8c CE	Segmented, drawn (?)	gB
XJ-16-12	Qiong-Ti-Mu site, Keping county	6–8c CE	Faceted, drawn (?)	G opaqe
XJ05-6-1	Da-Wang-Ku-Mu, Xinhe county	2c BCE–9c CE (Han to Tang)	Monochrome wound (?)	gB
XJ05-6-2	Da-Wang-Ku-Mu, Xinhe county	2c BCE–9c CE (Han to Tang)	Monochrome drawn (?)	gB
XJ05-6-3	Da-Wang-Ku-Mu, Xinhe county	2c BCE–9c CE (Han to Tang)	Monochrome	lgB
XJ05-6-4	Da-Wang-Ku-Mu, Xinhe county	2c BCE–9c CE (Han to Tang)	Monochrome	Y
XJ05-6-5	Da-Wang-Ku-Mu, Xinhe county	2c BCE–9c CE (Han to Tang)	Monochrome drawn (?)	lgB
XJ05-6-6	Da-Wang-Ku-Mu, Xinhe county	2c BCE–9c CE (Han to Tang)	Monochrome wound (?)	Bn
XJ05-6-7	Da-Wang-Ku-Mu, Xinhe county	2c BCE–9c CE (Han to Tang)	Stratified eye	pB, dG, W
XJ05-6-8	Da-Wang-Ku-Mu, Xinhe county	2c BCE–9c CE (Han to Tang)	Trailed decoration wound	gB, Bk
XJ-7	Shan-Pu-La cemetery, Luopu county	3–4c CE	Segmented silver-glass	Cls
XJ-39-1	Shan-Pu-La cemetery, Luopu county	1c BCE–4c CE	Monochrome drawn	V
XJ-39-2	Shan-Pu-La cemetery, Luopu county	1c BCE–4c CE	Monochrome drawn	V
XJ-40-1	Shan-Pu-La cemetery, Luopu county	1c BCE–4c CE	Monochrome drawn	R
XJ-40-2	Shan-Pu-La cemetery, Luopu county	1c BCE–4c CE	Mosaic eye Bara type	gB, R, Y, Bk
XJ05-2-1	Ke-You-Ke-Qin castle, Luntai county	2c BCE–2c CE (Han)	Folded and pierced (?)	gB
XJ05-2-2	Ke-You-Ke-Qin castle, Luntai county	2c BCE–2c CE (Han)	Monochrome drawn (?)	gB
XJ05-5-1	Red Flag Factory cemetery, Gongliu county	1c BCE–1c CE	Faceted, lapidary worked	aqua
XJ05-5-2	Red Flag Factory cemetery, Gongliu county	1c BCE–1c CE	Monochrome tubular	gB
XJ-5-1	Bao-Zi-Dong cemetery, Wensu county	4c BCE–2c BCE	Faceted (?)	pB
XJ-5-2	Bao-Zi-Dong cemetery, Wensu county	4c BCE–2c BCE	Monochrome drawn (?)	G
XJ05-8-1	Ma-Zha-Pu-Tang cemetery, Kuche county	2c BCE–2c CE (Han)	Faceted	gB
XJ05-8-2	Ma-Zha-Pu-Tang cemetery,Kuche county	2c BCE–2c CE (Han)	Monochrome drawn (?)	bG
XJ-21	Tuo-Gai-Ta-Mu castle site, Kuche county.	7–9c CE (Tang)	Monochrome wound ?	gB
XJ-23	Castle of Tang, Kuche county	7–9c CE (Tang)	Monochrome drawn (?)	G
XJ-34	Sen-Mu-Sai-Mu grotto site, Kuche county	later 2c CE to early 3rd C CE(Eastern Han Dynasty)	Stratified eye	gB, gB, W
XJ-12-3	Mai-Li-Ke-Ga-Wa-Ti Site, Hetian county	4–8c CE	Monochrome drawn	G
XJ-41	Mai-Li-Ke-Ga-Wa-Ti Site, Hetian county	4–8c CE	Monochrome drawn (?)	G
XJ-42-1	A'Ke-Si-Pi-Li castle ruins, Hetian county	2c BCE–13c CE (Han to Song)	Mosaic	Bk, gB, R, Y
XJ-42-2	A'Ke-Si-Pi-Li castle ruins, Hetian county	2c BCE–13c CE (Han to Song)	Stratified eye	Bk, gB, W
XJ05-4	Ku-Mu-Lu-Ke'ai-Ken cemetery, Baicheng	2c BCE–2c CE (Han)	Faceted wound (?)	gB
XJ05-7	A-He-Qi county	2c BCE–2c CE (Han)	Monochrome drawn (?)	lG
XJ-20	La-Yi-Su watch tower, Luntai county	10–13c CE (Song)	Monochrome	lG
XJ05-1	Tu-Mu-Xiu-Ke site, Bachu county	2c BCE–9c CE (Han to Tang)	Folded	Bk, W

Fig. 2. Among the samples, thirty-five were monochrome wound or drawn beads varying in largest dimension from a few millimetres up to 1 cm for some of the wound beads. An additional seven monochrome beads were faceted by various means including paddling the glass while hot, or grinding the bead surface following cooling. Four beads were drawn and then segmented; these include two 'gold-glass' beads with a thin metal foil, in one case silver (XJ-7) and in the other gold (XJ05-3-2), between two layers of glass. Polychrome beads were made by a variety of techniques, including six black wound beads with white trailed decoration and at least

Fig. 2. Glass beads excavated from Xinjiang, China.

three beads made by folding a trail-decorated glass pad around an iron mandrel, perhaps using a method suggested by Spaer (2001: 104). Two of these folded beads feature an additional wrap of either white or red glass (XJ05-1, XJ05-3-5). Two of the beads (XJ-42-1 and XJ-16-6) were made using a mosaic technique in which prepared polychrome elements, either slices of a mosaic cane or a pad of ribbon mosaic, were shaped into beads. In addition, there are two types of eye beads among the samples. Four are stratified eye beads, where a monochrome base bead (usually wound) was decorated with eyes built of layered pads of glass that were then marvered into the surface of the base bead (XJ05-3-7, XJ-42-2, XJ-34), or left as protruding eyes (XJ05-6-7). The other eye bead (XJ-40-2) was made with a different technique where slices of a mosaic eye cane were pressed into the surface of a monochrome base bead. While both stratified and mosaic eye beads were likely produced in several areas, the distinctive typology of this particular mosaic eye bead is identical to beads found in glassworking debris dated from the 1st c. BCE to the 2nd c. CE at Bara, a site in northeast Pakistan near Peshawar, at the time a capital of the Kushan Empire (Dussubieux and Gratuze, 2003). One surprising finding among the Xinjiang samples in this study was the presence of folded beads with trailed decoration, since these are usually considered to originate at Early Islamic sites of the 9th–11th century (Spaer, 2001).

3. Methods

3.1. Sample preparation

All the samples were weathered to different degrees. In order to decrease or eliminate the effect of surface weathering on the quantitative analysis, small areas of the sample surface were abraded using a diamond-coated fine grinding file to expose a fresh surface. The actual size of the polished area depended on the surface shape of the glass beads or fragments, but ranged between 25 and 55 mm^2. Our intention was to remove at least the outer 100 microns of weathered glass. Although, as mentioned above, this sample preparation was done in the laboratory, it could just as easily have been performed at an archaeological site or artefact storage area.

3.2. Analysis

An OURSTEX 100FA portable energy dispersive XRF (*pXRF*) spectrometer provided by OURSTEX Co., Ltd. was used for the chemical compositional analysis (Kato et al., 2010). A palladium (Pd) excitation tube was employed with voltage of 40 kV for monochromatic and filter modes or 15 kV for direct mode, and with a current of 0.5–1.0 mA. The detector is a silicon drift detector (SDD) with active area of 5 mm^2 equipped with a polymer window (MOXTEK AP 3.3 film), which can provide superior x-ray transmission in the low-energy range down to Be–Kα. The x-ray beam spot focused on the sample is about 2.5 mm in diameter. Together with the introduction of a vacuum sample chamber, this *p*XRF spectrometer allowed us to quantify the light elements such as Na and Mg. The SDD is cooled by a Peltier element, and operated at −28 °C. Elements from Na to Zn were determined by this spectrometer with the direct mode, and elements from Ag to Ba with filter mode. The other elements were determined with the monochromatic mode. The spectra are acquired for 300 s, 100 s for each of the three modes.

The data processing, including subtraction of background and calculation of the net peak areas, was done by the software package from OURSTEX Co., Ltd. The spectrometer is also equipped with a CCD system to confirm the area to be measured.

A total of twenty-two standard glass samples distributed by Breitlander Eichproben und Labomaterial GmbH, Germany, along with other synthetic glass samples, were used to make calibration curves for a number of major and trace elements in silicate glasses. Examples of the calibration curves for the quantitative analysis of the glass are shown in Fig. 3. A good correspondence between the experimental points and the regression lines was obtained for each element. The standard glasses NIST 1411 and SVF2, not used for the calibration, were measured 12 times to check the precision and accuracy of the method. The results are shown in the Table 2. This equipment shows good stability and the relative standard deviation (RSD) value for most of the major and minor elements detected is

less than 9%. The detection limits for some major and minor elements are also given elsewhere (Liu et al., 2011b). The bead sizes, shapes, surface roughness and various factors could affect the determined results in different degrees (Liu et al., 2011a), and normalized data are given in order to decrease these effects to the best of our abilities. We normalized the total of major and minor components in Table 3 to 100%.

4. Results

4.1. The affects of surface weathering on the analytical results

In order to demonstrate the affects of surface weathering, we selected thirteen beads, six soda glass and seven potash glass, to be analyzed on both the polished and weathered surfaces. Fig. 4 shows the resulting values for eight elements ranging in concentration from 0.2 to 20 wt%. Identity lines are shown in each graph, and the plotted values for weathered glass may be read as below or above those for the fresh glass surface. As expected, the measurements for soda (Na_2O) and potash (K_2O) (with one exception) are severely reduced in the weathered glass when the original content is above 5 wt%. On the other hand, when soda and potash levels are originally low, the measured values may actually increase, possibly due to absorption of alkali from the depositional environment. Alumina (Al_2O_3) content in weathered glass is almost always enriched, and such is the case with our study, where alumina may more than double. While lime (CaO) may decrease in weathered glass, it may also rise, as indicated in Fig. 4b. In our small sample, iron (Fe) may be enriched in unpredictable ways, while manganese (Mn) is the element least affected by weathering (Fig. 4c). This of course is at odds with commonly observed manganese increases in weathered glass, at times enough to cause a black crust to form. From Fig. 4d, titanium (Ti) may increase or decrease, resulting in levels considerably higher or lower than expected. While we have plotted the actual concentration differences, the values for percent change from expected would be even greater for the lower concentration ranges. Any use of such data for techniques depending on standardized values, such as principal components analysis (PCA) and cluster analysis, may lead to results to be interpreted with extreme caution.

Table 2
Measured and certified major and minor elements for samples NIST 1411 and SVF2. A total of 12 repeated analyses were carried out for each sample.

Sample	SV F2				NIST 1411			
Oxide	Average/ wt%	STD	RSD/%	Certified/ wt%	Average/ wt%	STD	RSD/%	Certified/ wt%
Na_2O	1.27	0.11	8.59	1.20	9.25	0.16	1.71	10.14
MgO	0.74	0.08	11.03	0.82	0.34	0.03	8.55	0.33
Al_2O_3	4.80	0.09	1.87	3.85	4.92	0.06	1.31	5.68
SiO_2	59.66	0.40	0.67	57.6	56.45	0.22	0.38	58.04
K_2O	18.50	0.19	1.02	18.4	3.20	0.03	1.02	2.97
CaO	2.81	0.05	1.74	2.84	2.05	0.01	0.70	2.18
TiO_2	0.10	0.01	13.38	0.04				0.01
V_2O_5	1.71	0.00	0.05	1.7				
Cr_2O_3	0.35	0.01	1.81	0.27				
Fe_2O_3	0.05	0.00	7.66	0.07	0.01	0.00	23.54	0.05
CoO	0.26	0.01	1.51	0.25				
CuO	1.99	0.02	1.14	1.8				
ZnO					3.86	0.02	0.43	3.85
Rb_2O	0.15	0.00	1.40	0.16				
SrO					0.08	0.00	1.05	0.09
Nb_2O_5	0.38	0.00	1.04	0.38				
CdO	0.97	0.01	0.59	0.96				
SnO_2	0.19	0.01	5.52	0.2				
Sb_2O_3	0.61	0.01	0.77	0.86				
BaO	0.24	0.04	16.81	0.34	5.15	0.15	2.89	5.00
PbO	0.01	0.00	25.46	0.05				

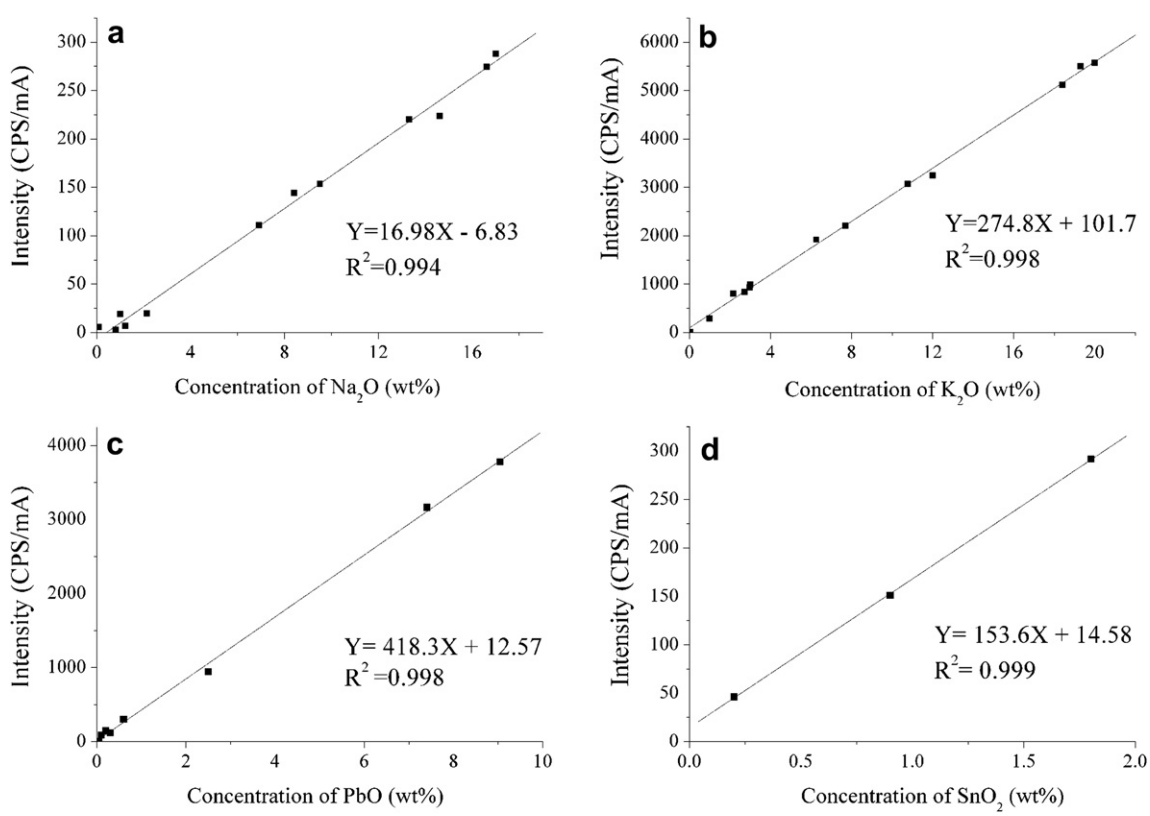

Fig. 3. Calibration curves for (a) Na_2O, (b) K_2O, (c) PbO, (d) SnO_2.

Table 3
Chemical compositional results for 88 different glasses from 65 beads found at various archaeological sites in Xinjiang, China. Major and some minor elements as oxides in weight percent (wt%); trace elements in parts per million (ppm). Abbreviations for colours as shown in Table 1.

Number	SiO_2	Na_2O	CaO	K_2O	N + K* norm	MgO	Al_2O_3	Fe_2O_3	Ti	MnO	CuO	Co	SnO_2	PbO	P_2O_5	Cl	Ba	Rb	Sr	Type
XJ-9-1	64.8%	16.9%	6.6%	3.0%	20.0%	3.1%	1.5%	0.9%	n.d.	0.1%	1.6%	n.d.	n.d.	n.d.	n.d.	1.5%	n.d.	60	510	v-Na–Ca SAS
XJ-9-2	65.2%	14.7%	8.8%	2.5%	17.2%	3.7%	1.4%	1.1%	263	1.5%	0.0%	n.d.	0.0%	0.1%	n.d.	1.0%	n.d.	38	348	v-Na–Ca SAS
XJ-9-3	62.9%	17.0%	5.6%	6.0%	23.0%	3.0%	1.7%	1.1%	444	0.1%	1.1%	n.d.	n.d.	n.d.	n.d.	1.3%	n.d.	71	294	v-Na–Ca Central Asia
XJ-9-4	60.2%	23.2%	6.0%	2.4%	25.6%	3.4%	1.8%	1.2%	688	0.1%	n.d.	n.d.	n.d.	n.d.	n.d.	1.6%	n.d.	55	576	v-Na–Ca SAS
XJ-10-1	72.0%	8.6%	5.5%	2.3%	11.1%	4.7%	1.5%	0.9%	n.d.	0.1%	1.4%	n.d.	1.1%	0.7%	n.d.	1.3%	n.d.	31	212	v-Na–Ca SAS
XJ-10-2	66.2%	10.5%	6.6%	2.0%	13.1%	3.4%	1.7%	1.1%	21	0.1%	2.4%	n.d.	1.1%	4.2%	n.d.	0.8%	n.d.	29	220	v-Na–Ca SAS
XJ-10-3	67.6%	11.2%	5.9%	3.3%	14.9%	2.7%	2.6%	1.0%	842	0.1%	1.6%	n.d.	1.0%	1.4%	n.d.	1.4%	n.d.	39	347	v-Na–Ca SAS
XJ-10-4	66.0%	17.3%	5.3%	4.0%	21.3%	1.9%	1.3%	0.8%	50	0.1%	1.9%	n.d.	n.d.	0.0%	n.d.	1.5%	n.d.	57	245	v-Na–Ca SAS
XJ-10-5	67.5%	12.4%	8.4%	2.1%	14.8%	4.1%	1.4%	1.3%	255	0.1%	0.1%	1920	0.6%	0.8%	n.d.	1.0%	n.d.	38	436	v-Na–Ca SAS
XJ-10-6	68.2%	9.3%	6.8%	2.6%	12.1%	4.1%	1.8%	1.3%	104	0.1%	2.4%	n.d.	0.4%	1.5%	n.d.	1.5%	n.d.	35	317	v-Na–Ca SAS
XJ-10-7	67.9%	13.6%	8.0%	2.7%	16.3%	3.7%	1.0%	0.9%	52	1.0%	n.d.	n.d.	n.d.	0.1%	n.d.	1.0%	n.d.	38	359	v-Na–Ca SAS
XJ-10-8	60.7%	23.3%	5.6%	3.2%	26.5%	2.5%	1.4%	1.4%	295	0.1%	0.2%	n.d.	n.d.	n.d.	n.d.	1.7%	n.d.	29	390	v-Na–Ca SAS
XJ05-3-1 Bk	64.2%	4.2%	6.0%	4.1%	9.0%	1.9%	4.6%	4.2%	358	1.0%	n.d.	n.d.	5.8%	2.2%	n.d.	1.8%	n.d.	38	224	v-Na–Ca Central Asia
XJ05-3-1 W	63.9%	8.3%	6.8%	5.1%	14.1%	2.7%	3.7%	1.8%	426	1.1%	n.d.	n.d.	3.9%	1.6%	n.d.	1.2%	n.d.	40	266	v-Na–Ca Central Asia
XJ05-3-2	71.5%	10.5%	10.1%	0.8%	11.3%	1.2%	3.3%	1.1%	248	0.1%	n.d.	n.d.	n.d.	0.0%	n.d.	1.3%	n.d.	50	889	m-Na–Ca natron
XJ05-3-3 Bk	62.0%	9.4%	6.7%	4.1%	13.6%	3.5%	2.4%	9.6%	493	0.6%	n.d.	n.d.	1.0%	0.1%	n.d.	0.7%	n.d.	43	215	v-Na-Ca Central Asia
XJ05-3-3 W	61.1%	11.9%	7.4%	4.4%	17.5%	3.8%	2.2%	1.4%	243	0.6%	n.d.	n.d.	3.2%	3.4%	n.d.	0.6%	n.d.	31	198	v-Na–Ca Central Asia
XJ05-3-4 Bk	61.1%	9.8%	8.1%	5.2%	15.0%	3.8%	3.0%	7.3%	505	0.1%	n.d.	n.d.	0.6%	n.d.	n.d.	0.9%	n.d.	48	265	v-Na–Ca Central Asia
XJ05-3-4 W	64.8%	5.0%	7.6%	4.1%	9.8%	3.2%	3.7%	3.1%	298	0.1%	n.d.	n.d.	2.5%	4.7%	n.d.	1.2%	n.d.	34	175	v-Na–Ca Central Asia
XJ05-3-5 Bk	59.8%	11.5%	7.4%	5.4%	16.9%	3.2%	2.8%	9.6%	822	0.1%	n.d.	n.d.	0.2%	n.d.	n.d.	n.d.	n.d.	53	204	v-Na–Ca Central Asia
XJ05-3-5 R	63.0%	7.3%	6.0%	3.5%	10.8%	3.0%	5.9%	7.8%	1315	0.1%	1.1%	n.d.	0.9%	0.3%	n.d.	1.0%	n.d.	45	269	v-Na–Ca Central Asia
XJ05-3-6 Bk	64.3%	9.0%	7.5%	4.6%	13.6%	3.5%	3.4%	6.3%	n.d.	0.3%	n.d.	n.d.	0.4%	0.1%	n.d.	0.7%	n.d.	48	261	v-Na–Ca Central Asia
XJ05-3-6 W	58.8%	5.2%	6.6%	3.2%	9.0%	2.2%	9.5%	2.7%	190	0.3%	0.1%	n.d.	3.2%	3.7%	n.d.	4.3%	n.d.	34	214	v-Na–Ca Central Asia
XJ05-3-7 B	63.3%	3.3%	6.0%	2.6%	6.5%	2.9%	6.8%	2.7%	1054	0.9%	0.0%	n.d.	8.5%	1.2%	n.d.	1.6%	n.d.	40	279	v-Na–Ca Central Asia
XJ05-3-7 Bk	57.7%	17.3%	4.7%	2.6%	20.7%	2.4%	4.4%	5.1%	1382	0.9%	n.d.	n.d.	3.9%	0.1%	n.d.	0.8%	n.d.	61	385	v-Na–Ca Central Asia
XJ05-3-8	55.9%	13.7%	4.4%	4.0%	17.8%	1.9%	8.5%	9.3%	1685	0.1%	n.d.	n.d.	0.5%	0.0%	n.d.	1.4%	n.d.	110	452	v-Na–Ca Central Asia
XJ-8-1	69.8%	13.4%	2.6%	3.1%	16.5%	0.9%	4.6%	1.3%	1611	0.2%	2.0%	n.d.	n.d.	n.d.	0.9%	0.6%	3426	116	207	m-Na–Ca-Al
XJ-8-2	71.3%	7.1%	4.3%	3.2%	10.3%	2.7%	3.5%	1.7%	896	0.1%	4.9%	n.d.	n.d.	n.d.	n.d.	1.2%	n.d.	46	253	v-Na–Ca Bara
XJ-16-1	64.9%	18.2%	3.3%	3.1%	22.0%	1.9%	1.7%	1.2%	300	0.1%	0.9%	n.d.	0.5%	2.5%	n.d.	1.6%	n.d.	46	173	v-Na–Ca SAS
XJ-16-2	70.2%	12.8%	4.9%	4.3%	17.1%	3.1%	1.4%	0.9%	n.d.	0.1%	1.1%	n.d.	n.d.	0.1%	n.d.	1.2%	n.d.	60	224	v-Na–Ca SAS
XJ-16-3	66.3%	15.0%	7.0%	3.1%	18.1%	5.2%	1.5%	1.0%	133	0.1%	n.d.	n.d.	n.d.	n.d.	n.d.	0.9%	n.d.	48	437	v-Na–Ca SAS
XJ-16-5	64.6%	19.1%	7.1%	3.1%	22.3%	3.4%	0.8%	0.6%	n.d.	0.1%	n.d.	n.d.	n.d.	0.0%	n.d.	1.1%	n.d.	38	342	v-Na–Ca SAS
XJ-16-6 gB	61.8%	17.0%	7.6%	2.1%	19.3%	3.2%	0.9%	0.5%	n.d.	0.1%	4.8%	n.d.	0.9%	0.4%	n.d.	0.8%	n.d.	34	314	v-Na–Ca SAS
XJ-16-6 Bn	66.7%	12.6%	9.0%	2.1%	14.8%	3.3%	3.0%	0.8%	n.d.	0.3%	n.d.	n.d.	0.9%	0.1%	n.d.	1.2%	n.d.	38	531	v-Na–Ca SAS
XJ-16-6 W	67.1%	11.6%	6.6%	1.2%	13.6%	3.2%	1.9%	0.7%	n.d.	0.4%	n.d.	n.d.	2.6%	3.4%	n.d.	1.3%	n.d.	36	209	v-Na–Ca SAS
XJ-16-7	68.0%	13.1%	7.1%	2.1%	15.2%	3.3%	3.1%	1.4%	290	0.1%	0.0%	n.d.	n.d.	0.7%	n.d.	1.0%	n.d.	35	341	v-Na–Ca SAS
XJ-16-8	66.7%	17.5%	2.4%	2.5%	20.0%	1.2%	4.3%	0.9%	1376	0.1%	3.3%	n.d.	n.d.	n.d.	n.d.	1.0%	n.d.	72	329	m-Na–Ca-Al
XJ-16-9	33.7%	1.1%	0.5%	2.3%	3.4%	0.4%	n.d.	0.1%	n.d.	n.d.	0.1%	n.d.	n.d.	61.6%	n.d.	0.3%	n.d.	29	25	PbSi
XJ-16-10 Bn	65.2%	6.9%	7.3%	2.4%	9.9%	4.0%	3.7%	1.0%	n.d.	0.3%	n.d.	n.d.	0.9%	5.6%	n.d.	2.7%	n.d.	34	339	v-Na–Ca SAS
XJ-16-10 Y	66.1%	6.7%	6.5%	2.1%	9.8%	3.3%	2.7%	0.8%	n.d.	0.4%	n.d.	n.d.	1.2%	8.8%	n.d.	1.4%	n.d.	32	182	v-Na–Ca SAS
XJ-16-11	61.1%	21.7%	5.1%	2.8%	24.7%	2.7%	1.7%	1.1%	322	0.1%	1.5%	n.d.	0.0%	0.6%	n.d.	1.5%	n.d.	50	328	v-Na–Ca SAS
XJ-16-12	61.4%	13.8%	6.0%	2.8%	17.5%	4.5%	1.8%	1.0%	418	0.1%	1.8%	n.d.	1.0%	4.7%	n.d.	1.0%	n.d.	28	234	v-Na–Ca SAS
XJ05-6-1	63.7%	21.7%	4.0%	2.6%	24.3%	2.1%	1.8%	1.1%	625	0.1%	1.1%	n.d.	n.d.	n.d.	n.d.	1.7%	n.d.	49	387	v-Na–Ca SAS
XJ05-6-2	66.9%	13.1%	5.5%	2.9%	16.0%	3.2%	2.4%	0.9%	85	0.1%	4.0%	n.d.	n.d.	n.d.	n.d.	1.0%	n.d.	70	240	v-Na–Ca SAS
XJ05-6-3	65.8%	19.0%	3.3%	3.2%	22.3%	1.0%	3.8%	0.9%	1062	0.1%	1.7%	n.d.	n.d.	n.d.	n.d.	1.0%	n.d.	80	344	m-Na–Ca-Al
XJ05-6-4	61.6%	9.9%	4.1%	1.3%	13.7%	3.5%	0.2%	0.4%	n.d.	0.1%	0.0%	n.d.	2.6%	15.9%	n.d.	0.5%	n.d.	34	99	v-Na–Ca SAS
XJ05-6-5	73.5%	2.3%	3.2%	14.4%	16.7%	0.7%	2.4%	0.5%	537	n.d.	1.6%	n.d.	n.d.	n.d.	0.8%	0.5%	n.d.	120	65	m-K–Ca–Al
XJ05-6-6	62.7%	16.1%	8.4%	3.8%	19.8%	4.7%	2.2%	0.9%	532	0.1%	n.d.	n.d.	n.d.	n.d.	n.d.	1.1%	n.d.	49	659	v-Na–Ca SAS
XJ05-6-7 pB	60.8%	7.9%	7.5%	5.2%	14.3%	3.3%	3.9%	1.7%	120	0.5%	n.d.	1383	7.2%	1.2%	n.d.	0.8%	n.d.	47	381	v-Na–Ca Central Asia
XJ05-6-7 dG	60.4%	9.5%	7.4%	5.2%	16.0%	3.2%	3.6%	1.5%	124	0.5%	n.d.	n.d.	7.1%	1.1%	n.d.	0.5%	n.d.	50	328	v-Na–Ca Central Asia
XJ05-6-7 W	50.2%	12.1%	9.2%	3.5%	17.0%	2.1%	1.7%	0.8%	n.d.	0.5%	n.d.	n.d.	6.9%	1.1%	n.d.	11.7%	n.d.	41	320	v-Na–Ca Central Asia
XJ05-6-8 gB	66.6%	3.4%	1.0%	n.d.	0.0%	2.3%	0.3%	0.2%	n.d.	n.d.	0.2%	n.d.	n.d.	21.5%	n.d.	2.3%	20,490	29	707	PbBa
XJ05-6-8 Bk	62.8%	2.9%	3.6%	0.0%	3.6%	2.6%	1.2%	4.8%	n.d.	n.d.	0.3%	n.d.	0.1%	17.5%	n.d.	1.9%	19,749	31	796	PbBa
XJ-7	75.3%	13.2%	4.2%	1.3%	14.5%	0.3%	1.6%	1.0%	369	1.7%	n.d.	n.d.	n.d.	n.d.	n.d.	1.5%	n.d.	38	350	m-Na–Ca natron.
XJ-39-1	78.7%	1.4%	1.4%	12.3%	13.7%	0.4%	1.1%	0.9%	1899	3.2%	n.d.	n.d.	n.d.	n.d.	n.d.	0.3%	n.d.	56	29	m-K–Ca–Al
XJ-39-2	78.7%	1.6%	2.3%	13.4%	15.1%	0.4%	0.5%	0.5%	551	2.2%	n.d.	n.d.	n.d.	n.d.	n.d.	0.4%	n.d.	51	21	m-K–Ca–Al
XJ-40-1	64.3%	8.5%	3.6%	0.1%	9.9%	0.9%	n.d.	0.7%	nd	0.3%	7.3%	0	0.2%	12.9%	0.0%	1.2%		29	52	m–Na-Ca–Pb–Cu (?)
XJ-40-2 Bk	63.2%	13.3%	4.9%	4.0%	17.3%	2.5%	3.3%	1.3%	562	0.1%	1.0%	n.d.	n.d.	4.9%	n.d.	1.1%	n.d.	39	294	v-Na–Ca Bara
XJ-40-2 gB	71.7%	13.7%	2.6%	2.4%	16.0%	1.0%	3.6%	0.8%	970	0.1%	2.9%	n.d.	n.d.	0.3%	n.d.	0.9%	n.d.	54	250	v-Na-Ca Bara
XJ-40-2 R	60.5%	14.7%	5.0%	4.1%	20.0%	2.6%	3.7%	1.3%	840	0.1%	1.0%	n.d.	0.4%	5.3%	n.d.	1.2%	n.d.	45	328	v-Na–Ca Bara
XJ-40-2 Y	67.6%	13.1%	3.5%	2.8%	16.9%	1.6%	2.6%	0.8%	518	0.1%	1.2%	n.d.	0.4%	5.2%	n.d.	1.0%	n.d.	39	167	v-Na–Ca Bara
XJ05-2-1	66.3%	15.0%	4.9%	4.0%	19.0%	2.2%	3.5%	0.9%	1788	0.2%	1.3%	n.d.	n.d.	n.d.	n.d.	1.2%	n.d.	72	408	v-Na–Ca Central Asia
XJ05-2-2	71.7%	10.8%	5.0%	3.3%	14.0%	1.8%	4.0%	1.1%	1412	0.1%	1.0%	n.d.	n.d.	n.d.	n.d.	1.0%	n.d.	74	314	v-Na–Ca SAS
XJ05-5-1	74.7%	6.9%	2.6%	8.7%	15.6%	1.3%	2.9%	0.8%	905	0.0%	n.d.	n.d.	n.d.	n.d.	1.3%	0.7%	n.d.	96	282	Mixed alkali
XJ05-5-2	65.1%	1.4%	2.8%	17.5%	18.9%	0.4%	2.2%	0.3%	221	0.0%	2.1%	n.d.	7.6%	n.d.	0.3%	0.2%	n.d.	112	30	m-K–Ca–Al
XJ-5-1	74.4%	0.9%	2.0%	17.3%	18.2%	0.5%	2.7%	0.4%	1044	0.0%	0.6%	532	n.d.	n.d.	0.7%	0.1%	n.d.	145	40	m-K–Ca–Al
XJ-5-2	75.5%	1.6%	2.2%	15.5%	17.2%	0.6%	1.8%	0.5%	511	0.0%	1.6%	n.d.	n.d.	n.d.	n.d.	0.5%	n.d.	141	38	m-K–Ca–Al
XJ05-8-1	72.8%	1.2%	2.4%	17.8%	19.0%	0.5%	2.4%	0.7%	840	0.0%	0.9%	n.d.	n.d.	n.d.	0.8%	0.4%	n.d.	145	45	m-K–Ca–Al
XJ05-8-2	75.8%	1.0%	2.5%	14.6%	15.6%	0.4%	2.4%	0.6%	789	0.0%	1.2%	n.d.	n.d.	n.d.	0.9%	0.4%	n.d.	151	41	m-K–Ca–Al
XJ-21	66.9%	14.3%	4.7%	2.4%	17.0%	5.5%	1.2%	0.5%	39	0.1%	1.7%	n.d.	0.8%	1.1%	n.d.	1.0%	n.d.	39	384	v-Na–Ca SAS
XJ-23	59.9%	16.6%	5.4%	3.1%	20.7%	2.6%	3.7%	1.9%	1136	0.1%	0.9%	n.d.	0.9%	4.1%	n.d.	0.8%	n.d.	49	272	v-Na–Ca SAS
XJ-34 B eye	71.3%	9.0%	6.2%	3.3%	12.8%	2.7%	0.9%	0.7%	n.d.	0.6%	0.6%	n.d.	3.7%	0.2%	n.d.	0.7%	n.d.	33	367	v-Na–Ca SAS

(*continued on next page*)

Table 3 (*continued*)

Number	SiO_2	Na_2O	CaO	K_2O	N + K* norm	MgO	Al_2O_3	Fe_2O_3	Ti	MnO	CuO	Co	SnO_2	PbO	P_2O_5	Cl	Ba	Rb	Sr	Type
XJ-34 gB	63.5%	15.7%	6.9%	3.2%	19.5%	3.8%	1.1%	0.5%	62	0.1%	1.2%	n.d.	2.2%	0.6%	n.d.	1.0%	n.d.	30	570	v-Na–Ca SAS
XJ-34 W	60.8%	8.9%	8.1%	3.2%	13.8%	3.3%	1.9%	0.7%	139	0.4%	n.d.	n.d.	9.8%	2.0%	n.d.	0.7%	n.d.	31	351	v-Na–Ca SAS
XJ-12-3	65.6%	12.3%	6.8%	1.5%	14.6%	3.5%	1.8%	1.0%	181	0.1%	1.2%	n.d.	0.9%	4.4%	n.d.	1.0%	n.d.	29	231	v-Na–Ca SAS
XJ-41	64.2%	8.6%	8.8%	4.3%	12.8%	4.5%	6.3%	2.1%	1265	0.1%	n.d.	n.d.	n.d.	n.d.	n.d.	1.0%	n.d.	76	562	v-Na–Ca Bara
XJ-42-1 Bk	65.8%	6.5%	9.2%	2.4%	9.3%	3.0%	3.3%	1.9%	941	0.4%	0.8%	n.d.	1.2%	1.9%	n.d.	3.3%	1687	34	331	v-Na–Ca SAS
XJ-42–1G, W	68.8%	6.2%	7.6%	2.7%	9.2%	3.5%	3.3%	1.3%	733	0.1%	2.3%	n.d.	1.3%	1.6%	n.d.	0.9%	1233	34	312	v-Na–Ca SAS
XJ-42-1 R	62.6%	7.8%	5.1%	4.8%	13.0%	3.1%	7.1%	1.6%	n.d.	n.d.	3.7%	n.d.	0.8%	2.0%	0.4%	1.0%	n.d.	39	281	v-Na–Ca SAS
XJ-42-1 Y	65.3%	4.6%	7.5%	2.7%	8.2%	2.4%	2.2%	1.3%	547	0.1%	1.9%	n.d.	1.9%	8.6%	n.d.	1.2%	1361	37	233	v-Na–Ca SAS
XJ-42-2 Bk	63.0%	4.8%	0.3%	n.d.	0.0%	0.2%	0.3%	7.9%	3181	0.0%	3.4%	n.d.	n.d.	17.0%	n.d.	1.9%	5907	29	47	PbBa (?)
XJ-42-2 gB	66.3%	6.2%	1.6%	n.d.	0.0%	0.5%	n.d.	0.4%	n.d.	n.d.	0.7%	n.d.	n.d.	19.4%	n.d.	1.6%	30,937	29	290	PbBa
XJ-42-2 W	77.5%	1.9%	n.d.	n.d.	0.0%	0.2%	0.2%	0.9%	n.d.	n.d.	0.5%	n.d.	0.0%	17.5%	n.d.	0.6%	6765	30	67	PbBa (?)
XJ05-4	73.0%	1.4%	2.2%	17.7%	19.1%	0.3%	2.3%	0.4%	837	0.0%	1.3%	n.d.	n.d.	n.d.	0.7%	0.4%	n.d.	156	36	m-K–Ca–Al
XJ05-7	79.5%	1.2%	2.1%	12.0%	13.2%	0.9%	2.3%	0.5%	655	0.0%	0.4%	n.d.	n.d.	0.1%	0.5%	0.4%	n.d.	112	34	m-K–Ca–Al
XJ-20	62.5%	13.5%	11.7%	3.4%	17.0%	3.4%	2.6%	1.4%	363	0.1%	n.d.	n.d.	n.d.	n.d.	0.2%	1.2%	n.d.	50	543	v-Na–Ca SAS
XJ05-1 Bk	56.0%	11.9%	7.1%	6.5%	19.2%	4.4%	5.6%	3.2%	411	0.3%	0.0%	n.d.	1.6%	2.6%	n.d.	0.7%	n.d.	71	245	v-Na–Ca Central Asia
XJ05-1 W	58.0%	10.9%	6.5%	5.0%	17.4%	3.5%	4.2%	1.1%	192	0.6%	n.d.	n.d.	2.9%	6.2%	n.d.	1.2%	n.d.	44	159	v-Na–Ca Central Asia

4.2. Glass chemical compositional types of the beads

The full analyses, based on polished surfaces, of 88 different glasses, including various colours for polychrome beads, for eighteen oxides or elements are given in Table 3, as weight percent (wt %) for major and some minor elements, and as parts per million (ppm) for trace elements. In addition, the sum of Na_2O and K_2O, normalized as if there were no lead (PbO) and tin (SnO_2), are included as N + K*, with the level indicating the degree of weathering for the soda and potash glasses, but not for the lead–barium (PbBa) and lead–silica (PbSi) samples. Our determination of the glass chemical compositional type for each sample is indicated as well, and will be explained below. Although we took pains to avoid weathered surfaces, about 10% of the individual

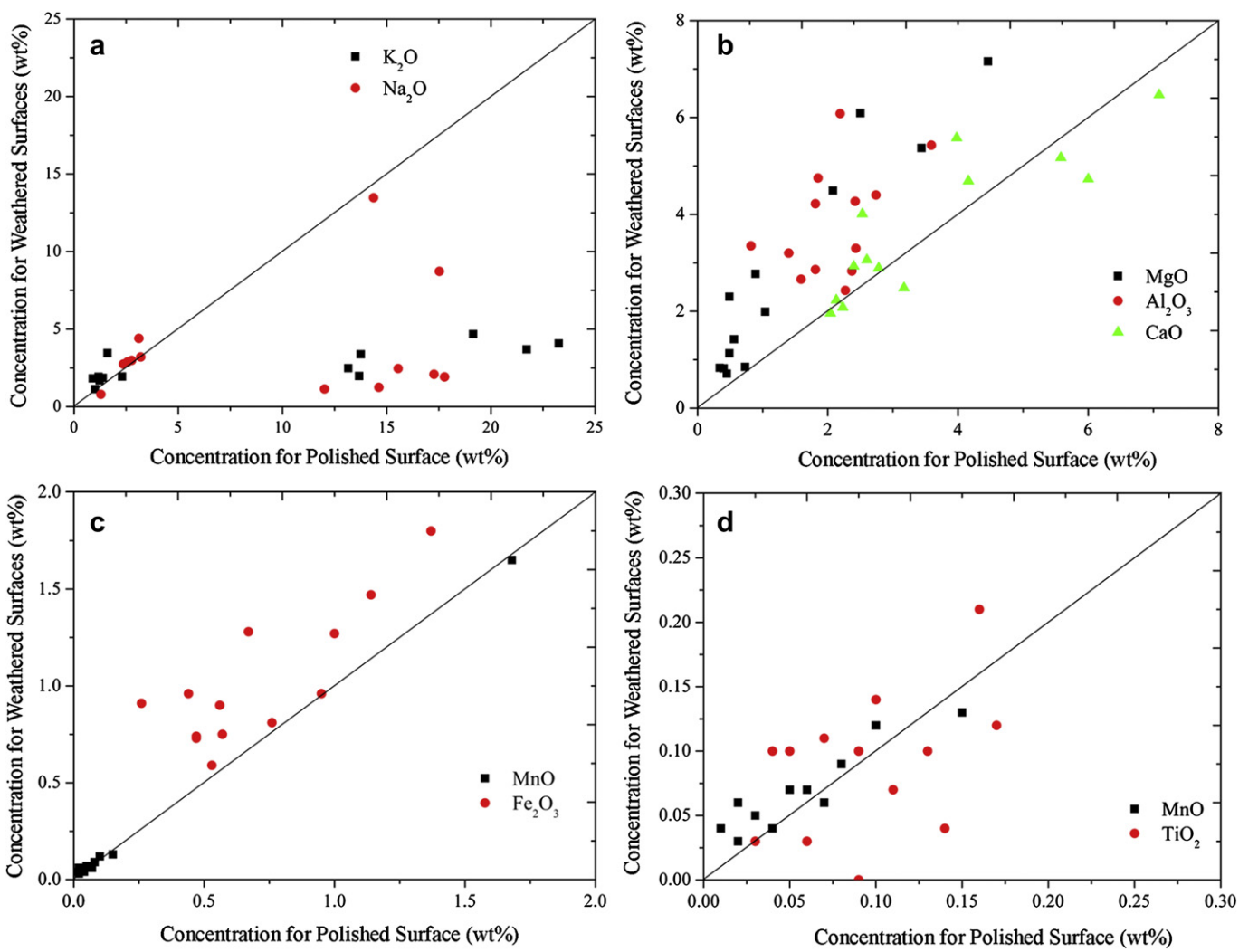

Fig. 4. Concentration comparisons for polished and weathered surfaces: (a) Na_2O and K_2O; (b) MgO, Al_2O_3 and CaO; (c) MnO and Fe_2O_3; (d) MnO and TiO_2.

analyses indicated moderately weathered glass, with alkali totals between 7 and 10 wt%. We included these analyses, often for single colour, in our determinations of glass type, taking into account that some values, particularly for alumina, might not be fully reliable.

There were seven glass compositional types in the 88 interpretable analyses representing 65 different beads. There was no good evidence for different chemical types being used for the same polychrome bead, so we will consider each bead to be of one chemical type. We have based these compositional types on major and minor elements rather than our measured trace elements, since most of the published comparison analyses did not include trace element determinations. The greatest number of samples, 45 beads or bracelets, were plant ash soda lime glass (v-Na–Ca, with v representing vegetal) with soda predominant over potash (Dussubieux, 2001), lime generally between 3% and 10 wt%, and magnesia greater than 1.5–2 wt%, indicating the use of a plant ash source for the alkali. We identified three different subtypes based primarily on levels of potash and alumina. The second largest group, 10 samples, was potash glass with low magnesia, indicating a mineral source for the potash, along with moderate levels of lime and alumina (m-K–Ca–Al, with m representing mineral) (Lankton and Dussubieux, 2006; Dussubieux and Gratuze, 2010). Beads from two groups of mineral soda lime glass were present. Three samples, including two gold–glass beads, had the low magnesia (MgO), moderate-to-high lime and low alumina characteristic of Mediterranean natron glass; the third sample that appears to have been made from natron soda glass (XJ-40-1), had a Raman spectrum (not shown here), indicating the existence of cuprite crystals. Three additional beads with lower lime and higher alumina may be termed m-Na–Ca–Al. Two beads were made from lead-barium (PbBa) glass characteristic for Chinese production. These included the stratified eye bead (XJ-42-2) and the wound and trail-decorated copper-blue bead (XJ05-6-8), both types characteristic for the Warring States to early Han Dynasty period. To complete the list, there were single examples of mixed alkali glass, with soda and potash both below 10 wt% but totalling 15 wt%, and lead-silica glass (PbSi) containing high levels of lead and silica and little else. The compositional types, along with their morphological, chronological and geographical characteristics, will be discussed further below.

5. Discussion

5.1. pXRF as a useful technique for the study of ancient glass

During the past few years, improvements in XRF technology have made relatively small, handheld, XRF spectrometers a reality. Measurements using such instruments have been helpful in the study of metals, particularly where invasive sampling is not possible. Unfortunately, this reality has been less promising for the area of glass study, for reasons that generally fall into two categories: instrumental and sample-dependent. Recently published papers illustrate both some of the possibilities and some of the problems. The first (Millhauser et al., 2011) compares the efficacy of a handheld *p*XRF spectrometer, operated without vacuum, to laboratory XRF on the same samples; the task is to provenance obsidian from central Mexico. Although the authors characterize the analytical results as 'inconsistent', with poor, or even negative, least-squares correlation between *p*XRF and laboratory XRF for individual samples, the difference observed between source areas was great enough that the *p*XRF values led to the 'correct' attribution of provenance in almost all cases. The quality of *p*XRF data from this comparison would likely be of very limited utility for the study of glasses of unknown types, both because of the inconsistent results stemming from the instrument itself, and because of the inability of atmospheric *p*XRF to detect elements lighter than K, thus eliminating Na, Mg, Al and Si, all very important for characterizing glass compositions. Additional observations were that adequate values could be obtained when the sample covered as little as 33% of the x-ray beam, and that concave surfaces gave poor results, presumably because of increased distance from the detector.

The second paper (Polikreti et al., 2011) applied *p*XRF to unprepared but relatively unweathered Mycenaean glass beads in order to distinguish an Egyptian or Mesopotamian origin for the glass. The authors depend on the suggestion by Shortland et al. (2007) that titanium (Ti) and zirconium (Zr) levels are elevated in Egyptian glass but not in Mesopotamian glass. These elevated Ti and Zr levels are thought to come not from the quartz pebbles used as a silica source, but possibly from the grinding stones used to prepare the quartz pebbles for melting (Rehren and Pusch, 2007), begging the question of whether everyone in Egypt used the same type of grinding stone. At any rate, the objects were all stylistically and archaeologically Late Bronze Age, and the authors make the relatively safe assumption that the unmeasured light elements would match those of typical Late Bronze Age glass. By plotting Ti and Zr values for their glass samples and comparing these to expected values for Egyptian and Mesopotamian glass, the authors were able to assign all but one of their samples to one category or the other. The *p*XRF apparatus used in this study appeared to work well to answer the authors' specific archaeological questions, in spite of the significant and unpredictable differences in Ti levels for weathered glass that both we and others have observed. For example, the analytical results from weathered and unweathered surfaces of some archaeological African glass beads reported by Dussubieux et al. (2009) using LA-ICP-MS showed that it is difficult to predict depletion or enrichment of each element, especially for trace elements, in the corroded layer. Many elements will behave differently depending on the type of corrosion. In addition, the heterogeneity of ancient glass will also affect the behaviour of each element in the corroded layer.

Kato et al. (2010) presented their results of chemical analysis of unpolished glass samples from Raya, a port on the southern coast of the Sinai Peninsula in Egypt, using an earlier version of the high-resolution *p*XRF spectrometer used for the present study. The samples were not polished, and measured soda levels were apparently low; those reported in the study were actually calculated by subtracting the sum of measured oxides from 100%. The quality of the other values seems to be good, since the results could be interpreted in terms of possible glass compositional types important during the period of activity at the site. It is possible that for these relatively lightly weathered samples from a desert environment only soda was severely affected; further work with such samples would clearly be of interest in helping us to understand the relationships of the various elements during the weathering process. For our samples from Xinjiang, all glasses were visibly weathered, and, for the most part, all elements were affected.

We believe that our own study has taken the use of *p*XRF on glass one step further. By combining the use of arguably the best available high-resolution portable XRF spectrometer with the potential to modify small areas of the glass surface, we have obtained analytical data for a group of samples of various compositional types that allows archaeological interpretation, as presented below.

Although not stressed in this report, the notion of portability lies at the heart of the advantages of *p*XRF for glass study. An increasing number of countries have followed Egypt's lead in making it very difficult to take samples abroad for analysis, so that even the best laboratory-based techniques are of no use if they are not available locally. In many parts of Asia, including China, this may be a problem as well. In addition, as archaeology becomes less

centralized, with the establishment of regional museums and archaeological institutions, even taking samples from one end of a large country to the other may be an issue, making a portable technique more advantageous.

5.2. Possible provenance of the glass beads

The temporal, geographical and morphological characteristics of the chemical types found in our study are listed in Table 4, and our

Table 4
Glass chemical compositional types by date, site, area (north or south of Taklamakan Desert), and sample number. For 'Han to Tang' and 'Han to Song' dates see Table 1.

Type	Date	Site	Area	Number
Early (4c BCE–4c CE)				
Mixed alkali	1c BCE–1c CE	Gonglin county, cemetery in Red Flag Factory	North central	XJ05-5-1
m-K–Ca–Al	4c BCE–2c BCE	Bao-Zhi-Dong, Wensu county	North central	XJ-5-2
m-K–Ca–Al	4c BCE–2c BCE	Bao-Zhi-Dong, Wensu county	North central	XJ-5-1
m-K–Ca–Al	Han to Tang	Da-Wang-Ku-Mu site, Xinhe county	North central	XJ05-6-5
m-K–Ca–Al	1c BCE–1c CE	Gonglin county, cemetery in Red Flag Factory	North central	XJ05-5-2
m-K–Ca–Al	2c BCE–2c CE	Ku-Mu-Lu-Ke'ai-Ken stone cemetery, Baicheng	North central	XJ05-4
m-K–Ca–Al	2c BCE–2c CE	Ma-Zha-Pu-Tang cemetery, Kuche county	North central	XJ05-8-2
m-K–Ca–Al	2c BCE–2c CE	Ma-Zha-Pu-Tang cemetery, Kuche county	North central	XJ05-8-1
m-K–Ca–Al	2c BCE–2c CE	A-He-Qi county	Northwest	XJ05-7
m-K–Ca–Al	1c BCE–4c CE	Shan-Pu-La cemetery, Luopu county	South	XJ-39-2
m-K–Ca–Al	1c BCE–4c CE	Shan-Pu-La cemetery, Luopu county	South	XJ-39-1
m-Na–Ca natron	2c BCE–4c CE	Niya	South	XJ05-3-2
m-Na–Ca natron	1c BCE–4c CE	Shan-Pu-La cemetery, Luopu county	South	XJ-7
m-Na–Ca natron (?)	1c BCE–4c CE	Shan-Pu-La cemetery, Luopu county	South	XJ-40-1
v-Na–Ca Bara	2c BCE–4c CE	Niya site, Minfeng county	South	XJ-8-2
v-Na–Ca Bara	4–8c CE	Mai-Li-Ke-Ga-Wa-Ti Site, Hetian county	South	XJ-41
v-Na–Ca Bara	1c BCE–4c CE	Shan-Pu-La cemetery, Luopu county	South	XJ-40-2
Probably early (1c BCE–4c CE)				
PbBa	Han to Tang	Da-Wang-Ku-Mu site, Xinhe county	North central	XJ05-6-8
PbBa	Han to Song	A'Ke-Si–Pi-Li castle ruins, Hetian county	South	XJ-42-2
Early (2c BCE–4c CE) with 1 possibly later sample				
v-Na–Ca Central Asia	2c BCE–2c CE	Niya	South	XJ05-3-4
v-Na–Ca Central Asia	2c BCE–2c CE	Niya	South	XJ05-3-3
v-Na–Ca Central Asia	2c BCE–2c CE	Niya	South	XJ05-3-1
v-Na–Ca Central Asia	2c BCE–2c CE	Ke-You-Ke-Qin castle, Luntai county	North central	XJ05-2-1
v-Na–Ca Central Asia	Han to Tang	Da-Wang-Ku-Mu site, Xinhe county	North central	XJ05-6-7
v-Na–Ca Central Asia	Han to Tang	Bachu county	Northwest	XJ05-1
v-Na–Ca Central Asia	2c BCE–4c CE	Niya	South	XJ05-3-8
v-Na–Ca Central Asia	2c BCE–4c CE	Niya	South	XJ05-3-6
v-Na–Ca Central Asia	2c BCE–4c CE	Niya	South	XJ05-3-5
v-Na–Ca Central Asia	2c BCE–4c CE	Niya	South	XJ05-3-7
v-Na–Ca Central Asia	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-9-3
Early and late (6–8c. CE)				
m-Na–Ca–Al	2c BCE–4c CE	Niya site, Minfeng county	South	XJ-8-1
m-Na–Ca–Al	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-8
m-Na–Ca–Al	Han to Tang	Da-Wang-Ku-Mu site, Xinhe county	North central	XJ05-6-3
Late (4c–9c CE) with 2 possibly early samples				
v-Na–Ca SAS	2c BCE–2c CE	Ke-You-Ke-Qin castle, Luntai county	North central	XJ05-2-2
v-Na–Ca SAS	Han to Tang	Da-Wang-Ku-Mu site, Xinhe county	North central	XJ05-6-1
v-Na–Ca SAS	Han to Tang	Da-Wang-Ku-Mu site, Xinhe county	North central	XJ05-6-6
v-Na–Ca SAS	Tang Dynasty	Tuo-Gai-Ta-Mu castle site, Kuche county	North central	XJ-21
v-Na–Ca SAS	Han to Tang	Da-Wang-Ku-Mu site, Xinhe county	North central	XJ05-6-4
v-Na–Ca SAS	Han to Tang	Da-Wang-Ku-Mu site, Xinhe county	North central	XJ05-6-2
v-Na–Ca SAS	Tang (618-907)	Castle of Tang, Kuche county	North central	XJ-23
v-Na–Ca SAS	Tang (618-907)	La-Yi-Su watch tower, Luntai county	North central	XJ-20
v-Na–Ca SAS	later 2c CE to early 3rd C CE	Sen-Mu-Sai-Mu grotto site, Kuche county	North central	XJ-34
v-Na–Ca SAS	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-6
v-Na–Ca SAS	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-10
v-Na-Ca SAS	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-11
v-Na–Ca SAS	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-2
v-Na–Ca SAS	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-7
v-Na–Ca SAS	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-3
v-Na–Ca SAS	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-5
v-Na–Ca SAS	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-1
v-Na–Ca SAS	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-12
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-9-4
v-Na–Ca SAS	Han to Song	A'Ke-SipPi-Li castle ruins, Hetian county	South	XJ-42-1
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-10-1
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-10-6
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-9-2
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-10-2
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-10-4
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-10-3

Table 4 (*continued*)

Type	Date	Site	Area	Number
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-10-7
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-10-5
v-Na–Ca SAS	4–8c CE	Mai-Li-Ke-Ga-Wa-Ti Site, Hetian county	South	XJ-12-3
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-9-1
v-Na–Ca SAS	4–8c CE	Da-Ma Gorge site, Cele county	South	XJ-10-8
Late (6c–8c CE)				
PbSi	6–8c CE	Qiong-Ti-Mu site, Keping county	Northwest	XJ-16-9

discussion will roughly follow this outline, beginning with the large number of plant ash soda lime glasses.

5.2.1. Plant ash soda lime glass (v-Na–Ca)

As mentioned above, the 45 beads or bracelets with a plant ash soda lime composition may be divided into three groups based primarily on the values for potash (K_2O) and alumina (Al_2O_3). In order to define these groups, we reviewed over 1000 published and unpublished analyses of plant ash soda lime glasses from Central Asia, Pakistan, Egypt, Syria–Palestine, Iran and Morocco (Table 5, with the percent of glasses from each site with K_2O and Al_2O_3 values above 4 and 4.5 wt%). As mentioned above, the alkali, including potash, in plant ash glasses is primarily introduced with the ash, although when sand high in feldspar is used as the silica source, additional alkali will be added. Plant ash itself may be quite variable in alkali content, depending on species of plant, growing conditions, the part of the plant that is collected and the temperature at which the plant is burned (Henderson, 2009; Barkoudah and Henderson, 2006). While such oxides as magnesia and lime show great variability within glasses from a particular site, even those thought to be areas of primary glass production, in our review the variations in potash (K_2O) appear to be more consistent, and thus more useful in distinguishing plant ash glass from different origins. Alumina, in contrast, comes primarily from the silica source. The plant ash itself contributes little, although clay incorporated with the plant material, as well as tools and containers used for making ash, crushing the quartz and melting the glass may have a variable, and difficult to quantify, contribution. In general, plant ash glasses with alumina less than 2.5 wt% are thought to have been made using crushed quartz pebbles as the silica source (Henderson, 2009), while sand, or a mixture of crushed quartz and sand, results in higher alumina values, in some cases up to 10 wt%. By comparing potash and alumina values of plant ash soda lime glasses from a variety of probable sources, we identified four subtypes of plant ash soda lime glass that could have been present in Xinjiang during the period covered by the sites in our study. The first of these is relatively low in both potash (less than 4.5 wt%) and alumina (less than 4 wt%), such as was found, and probably produced, at 9th to 13th c. sites in Egypt, western Asia, and Iran. Similar glass, at least in terms of these oxides, was produced in Sasanian-controlled Mesopotamia (Brill, 2005); we have called this plant ash subtype v-Na–Ca SAS, with SAS representing Sasanian. We have used the samples from Veh Ardashir, reported by Mirti et al. (2008, 2009) to represent this glass in Fig. 5a. Moving clockwise in Fig. 5a, the second group is a number of glasses found in northern Afghanistan analyzed by Brill (1999, 2009). Depending on the site, over half of these glasses have potash levels greater than 4 wt%, with almost as many greater than 4.5 wt%. Alumina values are variable, but usually less than 4.5 wt%. This group is represented in Fig. 5a by the samples from Aq Kupruk and Shahr-i-Banu. Because this group is currently defined by so few examples, for the moment we have called it v-Na–Ca Central Asia rather than the more precise 'Afghanistan'. There is clearly some overlap with the v-Na–Ca SAS glasses, but only for the relatively few v-Na–Ca Central Asia glasses with both lower K_2O and lower Al_2O_3. The third group, represented in Fig. 5a as 'Bara', also overlaps the previous two. While there is no evidence to show where this glass was made, the limited distribution, at least for the unworked glass, would suggest primary production in the relatively nearby area. For the Bara glass, the alumina values are most distinctive, with 80% of the analyses greater than 4.5 wt%, while the potash values are less diagnostic. However, the bead typologies from Bara were also distinctive, reflecting a Mediterranean technology of complex mosaic canes, mosaic eye beads, gold–glass beads and elliptical beads with combed blue and green longitudinal stripes (Dussubieux and Gratuze, 2003). The Bara material itself is relatively closely dated to one or two centuries before or after the turn of the first millennium CE. By combining chemical analysis, bead typology and chronology it is usually easy to distinguish Bara glass from Central Asian and Sasanian glasses in the few cases of overlap. Of course, it is also possible that some of the overlap may be explained by actual Bara glass at the Afghan sites, since these were not necessarily primary production sites and the glass samples were likely somewhat mixed. The fourth plant ash soda lime group was found at two sites in the Ferghana Valley in Uzbekistan (Osório, 2005; Rehren et al., 2010). That from Akhsiket is more likely to represent primary production, and is high in alumina but relatively low in potash, as shown in Table 5 and Fig. 5a. Although the Akhsiket samples were from the 9th to 12th c., glass with a similar composition has been reported from earlier sites in Uzbekistan (Abdurazakov, 2009). As a cautionary tale we included in Table 5 a number of glass samples found in Morocco, dated from the 12th to the 16th century. These are similar to the Akhsiket samples in terms of K_2O and Al_2O_3, but it seems very unlikely that they are in any way related. Rather, the Moroccan samples underline the importance of considering aspects of archaeology and chronology, as well as chemical similarity, in selecting comparison groups.

With four plant ash soda lime subtypes defined, we returned to our samples from Xinjiang and found a reasonable fit between their compositions and the defined groups. All of the monochrome plant ash soda lime Xinjiang beads from our study, whether drawn or wound, the faceted beads, segmented beads and the one zone mosaic bead fell into the lower K_2O and lower Al_2O_3 area defined by the Sasanian and Early Islamic glasses, as indicated in Fig. 5b. As stated above, this compositional subgroup is indicated as v-Na–Ca SAS in Tables 3 and 4. These glasses were found both north and south of the Taklamakan Desert, but are limited in date, with two possible exceptions (stratified eye bead XJ-34, monochrome bead XJ05-2-2), to the post-Han period beginning in the fourth century CE. In addition to most of the monochrome beads in the study, other beads with a v-Na–Ca SAS composition, and similar dating, include the mosaic bead (XJ-42-1) and a curious yellow bead that appears to have been decorated with brown trailing to imitate a mosaic cane bead (XJ-16-10). Why this particular type of plant ash soda lime glass appears only in the fourth century is not clear, but the simplest answer may be that the Sasanians were not making

Table 5
Xinjiang plant ash soda lime glasses from this study compared to 1102 plant ash soda lime samples from possible source areas by the percent of K_2O and Al_2O_3 values greater than 4.0 and 4.5 wt%.

Country	Site	Date	Samples	Reference	Number	$K_2O > 4$ wt%	$K_2O > 4.5$ wt%	$Al_2O_3 > 4$ wt%	$Al_2O_3 > 4.5$ wt%
China/Xinjiang	Various	Various	Beads	This study	64	30%	16%	14%	11%
Central Asia									
Afghanistan	Aq Kupruk	4c-Early Islamic	Vessels	Brill, 2009	7	57%	57%	29%	14%
Afghanistan	Sahr-i-Banu	7–13c	Vessels, bracelets	Brill, 1999	11	64%	55%	9%	0%
Afghanistan	Four sites	7–13c		Brill, 1999	8	25%	13%	25%	25%
Uzbekistan	Akhsiket	9–12c	Vessels, kiln glass	Osório, 2005; Rehren et al, 2010	32	19%	13%	63%	38%
Uzbekistan	Kuva	9–12c	Vessels	Osório, 2005; Rehren et al, 2010	11	55%	45%	91%	55%
Turkmenistan	Merv	Sasanian-Early Islamic	Vessels	Feuerbach, 2011	49	51%	14%	24%	24%
Turkmenistan	Merv	Early Islamic	Vessels	Feuerbach, 2011	29	45%	14%	45%	41%
China/Xinjiang	Kashgar (Haza-tam)	Prob. Roman or Sasanian		Brill, 2009	9	100%	100%	100%	89%
China/Xinjiang	Loulan	Various dates	Vessels, fragments	Brill, 2009	13	77%	46%	8%	0%
China/Xinjiang	Xinjiang Various sites	Various dates	Vessels ?	Brill, 2009	11	73%	55%	55%	36%
Central Asia totals					180	50%	29%	42%	32%
Pakistan	Bara	ca 1c CE	Beads, waste glass	Dussubieux and Gratuze, 2003; Dussubieux, 2001	59	61%	34%	86%	80%
Egypt Early Islamic									
Egypt	Fustat	8–12c	Beads	Lankton and Gratuze, unpublished	35	3%	0%	3%	3%
Egypt	Fustat	8–10+c	Vessels	Foy et al., 2003	30	3%	0%	20%	7%
Egypt	Fustat	9–13c	Vessels, cullet	Brill, 1999	29	0%	0%	0%	0%
Egypt	Raya, Wadi al-Tur	9–12c	Vessels	Kato et al., 2010	227	0%	0%	1%	0%
Egypt	various	10–12c	Glass weights	Gratuze and Barrandon, 1990	24	0%	0%	0%	0%
Egypt Early Islamic totals					345	1%	0%	3%	1%
Levant Early Islamic									
Lebanon	Tyre	10–11c	Raw glass, cullet	Freestone, 2002, Brill, 1999	11	0%	0%	0%	0%
Israel	Caesarea, Banias	9–13c	Vessels, chunks	Brill, 1999	32	0%	0%	0%	0%
Syria	Raqqa	9–11c	Raw glass, vessels	Henderson et al., 2004	164	1%	1%	4%	1%
Turkey	Serce Limani shipwreck	ca 1025	Vessels, blocks	Brill, 1999	103	0%	0%	2%	0%
Levant Early Islamic totals					310	1%	0%	3%	1%
Iran Early Islamic									
Iran	Nishapur, Gorgan, Hamadan	9–11c	Vessels	Brill, 1999	57	2%	2%	4%	0%
Iran	Siraf	Islamic	Vessels, waste glass	Brill, 1999	9	0%	0%	0%	0%
Iran Early Islamic totals					66	2%	2%	3%	0%
Sasanian									
China	Qizil	4–5c	Sasanian bowls	Brill, 2009	3	0%	0%	0%	0%
Iraq	Telll Umm Jirin	Sasanian ?	Vessels	Brill, 2005	6	0%	0%	33%	17%
Iraq	Choche (V. Ardashir)	3–5c	Vessels, waste glass	Brill, 2005	11	36%	9%	0%	0%
Iraq	Jezaziyat	Sasanian-Early Islamic	Vessels	Brill, 2005	21	10%	0%	0%	0%
Iraq	Tulul Umm Ghemimi	2–6c	Vessels, cullet	Brill, 2005	12	0%	0%	0%	0%
Iraq	Veh Ardashir	3–6c	Vessels and cullet	Mirti et al, 2008, 2009	71	10%	1%	0%	0%
Sasanian totals					124	10%	2%	2%	1%
Morocco	Qsar-es-Seghir	12–16c	Vessels	Brill, 1999	18	22%	0%	67%	33%

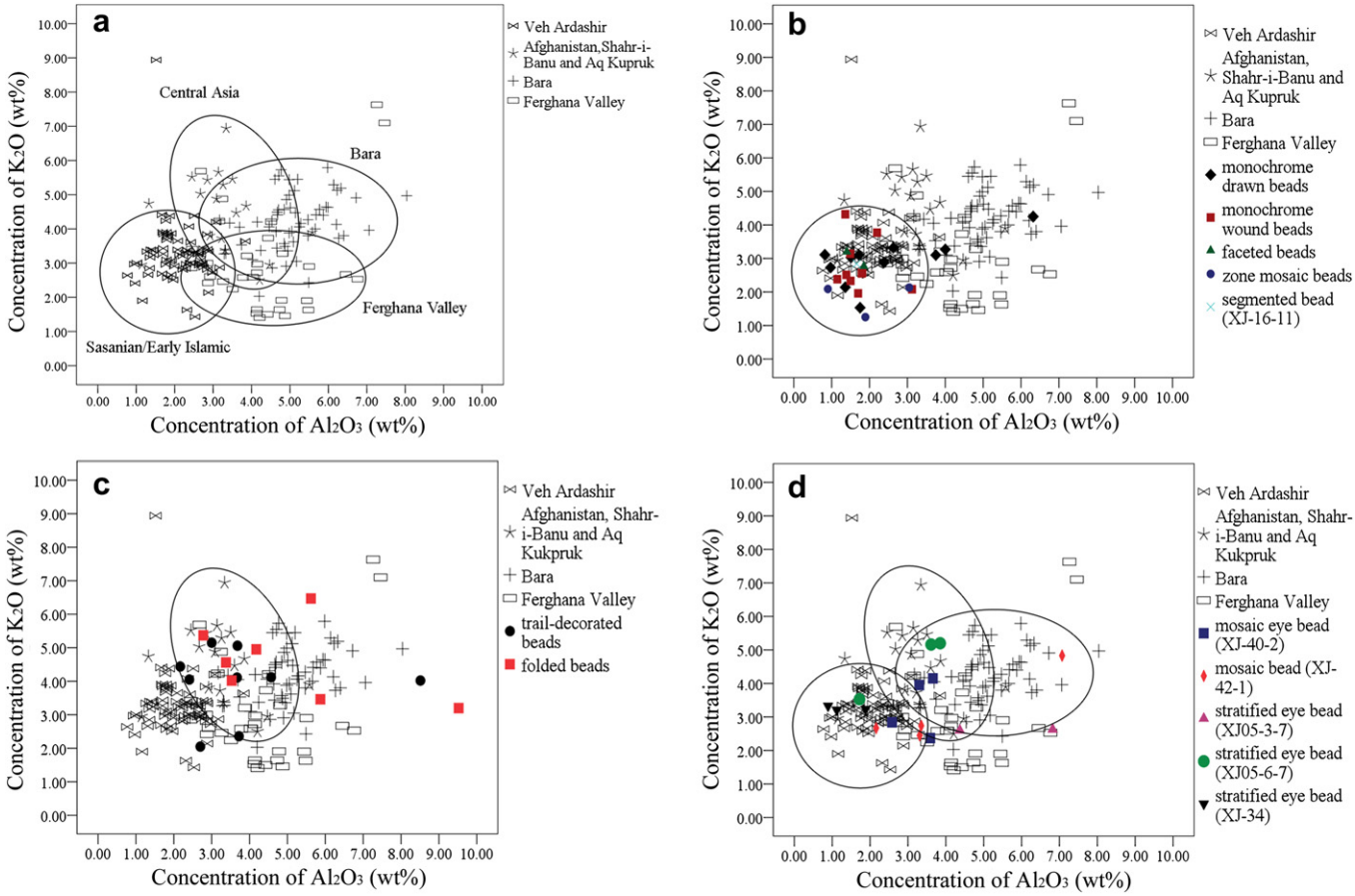

Fig. 5. (a) Possible source areas (K_2O *vs* Al_2O_3); (b) Xinjiang monochrome drawn and wound beads, segmented beads, faceted beads and zone bead (K_2O *vs* Al_2O_3); (c) Xinjiang folded and trail-decorated beads (K_2O *vs* Al_2O_3); (d) Xinjiang eye and mosaic beads (K_2O *vs* Al_2O_3). Ellipses are added only to indicate glasses from the individual source areas.

much glass prior to the fourth century, a supposition supported by the dates from the Sasanian glass production site of Veh Ardashir (Mirti et al., 2008, 2009), as well as by the appearance of similar plant ash soda lime glass beads in both Southeast Asia (Lankton and Dussubieux, 2006) and Korea (Lankton and Lee, 2006) at about this same time. Relatively little is known about glass bead production in the Sasanian Empire. In Simpson's (2003: 68) review of the available evidence, he points out that glass beads found at Sasanian sites tend to be furnace-wound rather than drawn, although the presence of Indian communities at such sites as Ubulla, the Sasanian precursor of Basra, could explain the beads made with Sasanian plant ash soda glass by a drawing technique, typically the purview of Indian craftsmen, that have been found in Xinjiang (this study), Southeast Asia and Korea.

All of the trail-decorated and folded beads (Fig. 5c) from our study are consistent in chemical composition with the Afghan samples defining the 'Central Asia' portion of our graph. Most of the apparent exceptions are individual colours from polychrome glass beads, where the other colours had a more typical Central Asian composition, or where the weathered nature of the glass makes higher than expected alumina values less surprising. The plant ash soda lime glasses from our study with a Central Asian composition are noted as v-Na–Ca Central Asia. From Table 4, the v-Na–Ca Central Asia samples were found both north and south of the desert, and, with one possible exception, all of these glasses are early. While it is easy to imagine early glass production in both northern Pakistan and Bactria (northern Afghanistan bounded by the Oxus or Amu Darya River), it is more surprising that such glasses are not found at sites post-dating the fourth century CE. Based on current information, it is difficult to specify whether glass beads with a Central Asian composition were the private possessions of the local populations, or whether they were a commodity of trade. Either way, the apparent decrease in beads made from v-Na–Ca Central Asia glass at the same time that beads with a typical Sasanian composition became important, may represent changing patterns in glass production, changing patterns of trade, changing populations in the oasis centres, or perhaps a bit of all three. Close attention to glass and other artifacts at the sites around the Taklamakan, coupled with a better understanding of what was happening in both glass source areas and more distant glass consumption areas, may help to answer such questions.

One additional comment about the morphology of the 'Central Asian' glass beads: As mentioned above, polychrome folded glass beads have been considered to be an invention of beadmakers working at Early Islamic sites in the 9th–11th century. While it is conceivable that at least some of the folded beads from our study may be later intrusions, their consistent dating, and, in addition, the consistent dating of the v-Na–Ca Central Asia glass, makes this less likely. Rather we suggest that such beads were produced at Central Asian sites long before they appear in western Asia, and folded beads can no longer be assumed to have an Early Islamic provenance.

Possible Bara glass, representing an early Kushan type, is represented in our study by three examples: the mosaic eye bead (XJ-40-2), a segmented drawn bead (XJ-8-2) and a monochrome bead (XJ-41) that appears to have been drawn. Eye beads with an identical morphology were found at Bara, as well as the previously mentioned example from Niya (Lin, 2010), and segmented beads

are characteristic for Bara production, based on photographs published by Dussubieux and Gratuze (2003). The chemical compositions for our beads are a good match with those from Bara reported by Dussubieux (2001), even though the alumina values for our mosaic eye bead fall into the lower range for Bara glass. The presence of these samples along the southern limb of the oasis routes of the Silk Road is consistent with known Kushan influence in this area. At least some of these beads were likely to have been personal possessions of Kushan people settled in or travelling through the Xinjiang oases urban sites.

5.2.2. Potash glass with moderate lime and alumina (m-K–Ca–Al)

The ten potash glass beads from our study were all monochrome and mostly drawn. The values for major elements for these glasses are very similar to those for potash glass samples from Arikamedu in southern India and at the North Indian site of Kausambi (both datasets Lankton and Gratuze, unpublished), as well as many of the potash glasses found in southern China (e.g., Shi et al., 1986; Huang, 1991; Li et al., 2009), Southeast Asia (Basa et al., 1991; Lankton et al., 2008), and Korea (Lankton and Lee, 2006). Where this glass was produced is not clear, although multiple production areas seems likely given the widespread distribution, and the extent to which the m-K–Ca–Al glass is 'Chinese' is an important question. It is likely that detailed analysis of trace elements, as well as possibly strontium and neodymium isotope studies, will be necessary to tease out the likely subtypes of m-K–Ca–Al glass. From our own data, while it is possible that all of the Xinjiang samples came up through northern India, during the first and second centuries CE a part of the Kushan Empire, the actual distribution at our Xinjiang sites is different from that of the Bara samples, although admittedly the sample numbers are low. While the Bara glasses are all found on the southern limb of the desert routes, the potash glass is mainly northern, suggesting a different distribution mechanism and possibly origin. This difference is certainly a topic for further study.

5.2.3. Soda glass with moderate-to-high lime and low magnesia

The Mediterranean ('Roman') natron m-Na-Ca gold-glass beads (XJ-7, XJ05-3-2) represent a special class of high value ornaments, and have been found as far afield as Southeast Asia and Korea (Lankton, 2009), even when evidence for other Mediterranean glass is scarce. Although both of our samples were from sites on the southern limb of the oasis routes, this may be a function of the few beads analyzed.

5.2.4. Mineral soda lime glass with high alumina (m-Na–Ca–Al)

Our samples include three beads with an m-Na–Ca–Al composition, low in magnesia, but lower in lime and higher in alumina than Mediterranean natron glass. When trace elements are available, this glass may be distinguished by higher levels of uranium (Dussubieux, 2001). India or Southeast Asia is the most likely production centre, and glass similar to that found in Xinjiang is also present at Kausambi, suggesting a route to Xinjiang via northern India. This m-Na-Ca-Al glass was probably produced over a long period of time, consistent with our Xinjiang data.

5.2.5. Lead-barium glass (PbBa)

The two samples of lead-barium glass from our Xinjiang beads are both morphologically typical for Warring States to early Han Dynasty production, and such an early date is consistent with the dating where these beads were found (both sites covering a long period but including early material). To date, such beads have not been found further west, so we suspect that these particular beads represent Han Chinese people's presence in Xinjiang or the mutual exchange between Xinjiang and central China during the early period. Further study of material from Xinjiang may help to better define the meaning of these important beads.

It is difficult to be much more specific about the remaining types of glass, each represented by single beads found in our study, although at least the mixed alkali glass has been associated with Central Asian production (Brill, 2009). The lead-silica glass may be Chinese, since the primary production of similar glass in Korea and Japan (Lee, 1994; Yamasaki and Murozumi, 1991) was possibly influenced by imported Chinese technology.

6. Conclusions

In response to our original archaeological questions, we can now answer that glass beads with at least seven different chemical compositional types were present at our eighteen different sites in Xinjiang, with good correlation between chemistry, morphology and chronology for most of the beads. Early (2nd c. BCE–4th c. CE) glass types include both Central Asian and Kushan variants of plant ash soda lime glass, with a distribution for the latter centred on the southern limb of the oasis Silk Routes. Potash glass is also uniformly early, and found both north and south of the Taklamakan Desert, raising questions of its ultimate sources and pathways to Xinjiang. High value Roman glass, in the form of gold-glass beads, is present, either as prized possessions of oasis residents, or perhaps on its way to distant markets in Korea, since relatively few samples have been reported from central China. Based on compositional type, the beads made from mineral soda glass with moderate lime and high alumina were almost certainly Indian or Southeast Asian, and may have entered Xinjiang via Kushan-held territory in northern India. The two lead-barium glass beads studied represent Chinese technology. Whether these belonged to Han Chinese people is not clear, although ancient ornament styles were often very traditional and closely linked to particular cultures. Because of this, we may cautiously suggest that, based on the available evidence for glass chemical compositions and morphological types from the early period, the number of Han Chinese buried at these Xinjiang sites was rather limited. During the later period (4th–10th c. CE) the glass compositions change dramatically with the near disappearance of both potash glass and Central Asian plant ash glass, being replaced by a new type of plant ash glass more similar to that produced at Sasanian sites in Mesopotamia. Although at least some Sasanian people were probably present in Xinjiang at this time, we consider it likely that this Sasanian glass was more an object of trade, perhaps brought by the rising group of Sogdian traders that would come to dominate the Central Asian Silk Routes.

In terms of our methodological question of how best to combine quantitative, on site, analysis with a minimally destructive technique, we believe we have found a way forward. The data from this pilot study was more than adequate to allow an archaeological interpretation of the results, and fits well with, while expanding upon, previous analyses done with more traditional techniques. In addition, the marks left on the surface of the samples, although visible, did not alter the shape or meaning of the glass remains. Further steps will be to refine the sample preparation methods, possibly using smaller polished areas, and to do a more direct comparison of *p*XRF data for archaeological samples with that obtained by more standard techniques such as electron probe microanalysis (EPMA) and laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS). Such a study has already been done by the group at Tokyo University of Science using an earlier version of their *p*XRF equipment, with promising results (Tantrakarn et al., 2011), but would bear repeating for glass of different chemical compositional types and different depositional circumstances.

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Appendix. Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jas.2012.02.035.

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